

National University of Computer and Emerging Sciences
Lahore Campus

Artificial Intelligence Lab

(AL2002)

Date: June 3rd, 2026

Course Instructor(s)

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Final Exam

Total Time (Hrs): 2hrs

Total Marks: 50

Total Questions: 2

Roll No

Section

Student Signature

Do not write below this line

Attempt BOTH questions.

Drag and Drop the file into your respective submission folder. The folder will close automatically on time, hence submission on time, submission of correct .ipynb file with your rollNo is your responsibility. If you fail to do so, you will be awarded with zero marks.

Submission Path: \\exam\Exam\Final Exam\Artificial Intelligence\Your Section\Submissions

Dataset Path: \\exam\Exam\Final Exam\Artificial Intelligence\Your Section \Paper

Q1: CNN

[28 marks]

Imagine you work as an AI engineer in a beverage factory. Your task is to build a computer vision model that automatically checks juice boxes on the production line and classifies defects before the products are shipped to customers

- a) Load the given dataset from the directory structure using `image_dataset_from_directory` from tensorflow (3)

```
import tensorflow as tf

IMG_SIZE = (128, 128)

train_dataset = tf.keras.preprocessing.image_dataset_from_directory(
    "dataset/train", #path to the folder containing the training images
    image_size=IMG_SIZE
)

val_dataset = tf.keras.preprocessing.image_dataset_from_directory(
    "dataset/valid", #path to the folder containing the validation images
    image_size=IMG_SIZE
)
```

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```
test_dataset = tf.keras.preprocessing.image_dataset_from_directory(
    "dataset/test", #path to the folder containing the test images
    image_size=IMG_SIZE,
    shuffle=False # Important for evaluation
)

class_names = train_dataset.class_names
print("Classes:", class_names)
```

- b) Build a CNN using tensorflow/keras with the following layers: (8)
- a. Input Layer
 - b. Convolution Layer (32 filters, 3×3, ReLU)
 - c. MaxPooling Layer (2×2)
 - d. Convolution Layer (64 filters, 3×3, ReLU)
 - e. MaxPooling Layer (2×2)
 - f. Flatten Layer
 - g. Dense Layer (64 neurons, ReLU)
 - h. Output Layer
- c) Compile the model with: optimizer='adam', loss='sparse_categorical_crossentropy' (2)
- d) Train the model using your dataset with **3-fold cross validation** and the following parameters: epochs=15, batch_size=32 (2+3)
You would need to convert the dataset to numpy arrays to apply k-fold cross validation. Use the code given below:

```
X = []
y = []
for images, labels in train_dataset:
    X.append(images.numpy())
    y.append(labels.numpy())
X = np.concatenate(X, axis=0)
y = np.concatenate(y, axis=0)
```

- e) Train the same model again with same parameters and **3-fold cross validation** but with **early stopping** (patience=3) (2)
- f) Compare the performance of both models using mean cross validation accuracy. Which model will you choose for final classification? (2+1)
- g) Evaluate the performance of your final chosen model on test data by making use of: (4)
- a. Classification Report
 - b. Confusion Matrix
- h) Is your chosen model underfitting, overfitting or well-fitted? (1)

Q2: Regression

[22 marks]

Imagine you work as a data analyst in a real estate company. Your task is to predict house prices based on the size of houses and number of bedrooms using machine learning models. The following data is provided:

House Size (sq ft)	Number of Bedrooms	Price (\$1000)
800	2	150
1000	2	180
1200	3	210
1400	3	240
1600	4	270
1800	4	300
2000	5	330
2200	5	360

Part A

[7 marks]

- a) Create a pandas DataFrame from the given data. (2)
- b) Separate the dataset into Features (X) and target label (y) (2)
- c) Split the dataset into 20% test and 80% training set using the code below: (1)

```
X_train, X_test = train_test_split(df, test_size=TEST_PERCENTAGE, random_state=42)
```

- d) Standardize the dataset. (2)

Part B

[4 marks]

- e) Train a Linear Regression model (LinearRegression()) from scikit-learn. (2)
- f) Evaluate the performance of your test dataset on this model using Mean Squared Error (MSE). (2)

Part C

[11 marks]

- g) Using this dataset, Implement a Single Layer Perceptron *from Scratch*. Your implementation should include:
 - a. A training function iterates over the dataset to update weights and bias for 30 epochs with a learning rate of 0.1. This function should return trained weights and bias (5)

$$w_i = w_i + \Delta w_i \quad \Delta w_i = \eta \cdot (y - \hat{y}) \cdot x_i \quad b = b + \eta \cdot (y - \hat{y})$$

Note: Since this is a regression problem, there will be no activation function.

- b. A predict function that computes the output for test data using the trained weights and bias. (2)

$$z = \sum_{i=1}^n w_i x_i + b$$

- c. Extend your implementation to allow a user-provided test input (a single new data sample) and output the prediction. Demonstrate this functionality with an example input. (2)
- h) Evaluate the performance of your test dataset on this model using Mean Squared Error (MSE). Which model performs better? (1+1)

Reference Table for Libraries

Main Library	Sub-Libraries / Classes / Functions
<code>sklearn.preprocessing</code>	LabelEncoder, OrdinalEncoder, StandardScaler, MinMaxScaler
<code>sklearn.model_selection</code>	train_test_split, KFold, cross_val_score
<code>sklearn.metrics</code>	accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, classification_report, mean_squared_error, mean_absolute_error, r2_score
<code>sklearn.linear_model</code>	LinearRegression
<code>sklearn.cluster</code>	KMeans
<code>tensorflow.keras.models</code>	Sequential, Model
<code>tensorflow.keras.layers</code>	Dense, Input, Conv2D, MaxPooling2D, Flatten, Dropout,
<code>tensorflow.keras.callbacks</code>	EarlyStopping, ModelCheckpoint, ReduceLROnPlateau
<code>tensorflow.keras.optimizers</code>	Adam, SGD, RMSprop

Good Luck!